

| Model              | Method                  | Cancer-Myth       |                   | CREPE |       |
|--------------------|-------------------------|-------------------|-------------------|-------|-------|
|                    |                         | TPR ↑             | FPR ↓             | TPR ↑ | FPR ↓ |
| Qwen2.5<br>7B      | Direct gate             | 7.0 <sup>H</sup>  | 10.7 <sup>H</sup> | —     | —     |
|                    | Premise-review CoT gate | 29.5 <sup>H</sup> | 16.1 <sup>H</sup> | —     | —     |
|                    | GEPA                    | —                 | —                 | —     | —     |
|                    | GEPA + text feedback    | —                 | —                 | —     | —     |
|                    | GEPA + SAE feedback     | —                 | —                 | —     | —     |
| GPT-6<br>Luna      | Direct gate             | —                 | —                 | —     | —     |
|                    | Premise-review CoT gate | —                 | —                 | —     | —     |
|                    | GEPA                    | —                 | —                 | —     | —     |
|                    | GEPA + text feedback    | —                 | —                 | —     | —     |
|                    | GEPA + SAE feedback     | —                 | —                 | —     | —     |
| Gemma 4<br>12B     | Direct gate             | 40.5 <sup>H</sup> | 16.1 <sup>H</sup> | —     | —     |
|                    | Premise-review CoT gate | 51.3 <sup>H</sup> | 32.9 <sup>H</sup> | —     | —     |
|                    | GEPA                    | —                 | —                 | —     | —     |
|                    | GEPA + text feedback    | —                 | —                 | —     | —     |
|                    | GEPA + SAE feedback     | —                 | —                 | —     | —     |
| Qwen3.8<br>27B OFF | Direct gate             | 26.1 <sup>H</sup> | 16.1 <sup>H</sup> | —     | —     |
|                    | Premise-review CoT gate | 57.5 <sup>H</sup> | 35.6 <sup>H</sup> | —     | —     |
|                    | GEPA                    | —                 | —                 | —     | —     |
|                    | GEPA + text feedback    | —                 | —                 | —     | —     |
|                    | GEPA + SAE feedback     | —                 | —                 | —     | —     |
| Qwen3.8<br>27B ON  | Direct gate             | 44.1 <sup>H</sup> | 21.5 <sup>H</sup> | —     | —     |
|                    | Premise-review CoT gate | 61.4 <sup>H</sup> | 38.3 <sup>H</sup> | —     | —     |
|                    | GEPA                    | —                 | —                 | —     | —     |
|                    | GEPA + text feedback    | —                 | —                 | —     | —     |
|                    | GEPA + SAE feedback     | —                 | —                 | —     | —     |

**Table 1. False-premise detection across models and datasets**

Stored results; — means unavailable or not comparable. Values are percentages, one run.

TPR ↑: FPQ detection; FPR ↓: false positives on normal questions.

H: historical full-corpus evaluation (583 FPQ / 149 NFP); valid denominators in ledger.

Model versions, counts, judge/protocol differences and source hashes: result\_sources.md.